Forgetting to Refuse, or Forgetting to Freeze? A Taught VLA Safeguard Under Benign Personalization

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Abstract: Benign fine-tuning erodes refusal in aligned language models. Does a refusal taught to a vision-language-action (VLA) policy survive benign personalization? In two pre-registered studies we teach OpenVLA-7B to hold still when told to harm a human hand, personalize it with LoRA on 200 benign demonstrations, and compare its closed-loop hand contact with that of a comparator never taught to refuse. We find that the taught refusal worked mostly by freezing: the policy held still under benign as well as harmful instructions. Personalization loosened that freeze under ordinary instructions, and much of its protection went with it: inside the personalization domain, hand contact rose in all three seeds (seed 0: 0 to 40 of 150 episodes), and in seed 0 the policy was no longer detectably safer than the comparator. A narrower stop specific to the harmful instruction did survive: in post hoc runs, the personalized policy stayed still far more often when told to harm the hand than when told to avoid it (148 vs. 97 of 150 episodes, seed 0); in two further seeds, tested against the plain task only, the gap was smaller and did not meet our declared replication criterion. Whether that stop would prevent harm could not be tested, because no policy, aligned or not, detectably carried out the harmful instruction. We distill this into three traps for evaluating VLA refusal, each with a fix.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], and alignment is often shallow and its durability overstated [3, 4]. Safety is now installed in the weights of VLA policies [5] and measured in closed loop [6], but each safeguard is trained and evaluated once, whereas a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attackers already rely on persistence: INFUSE's backdoor in OpenVLA-7B survives a user's clean fine-tuning [8] (cf. BadVLA [9]). We ask the defender's question, which task retention does not answer, since VLAs forget heavily under the plain supervised fine-tuning that personalization uses [10, 11].

VLAs raise two measurement problems that text models do not. Refusal is a no-op action while harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to respond to the instruction, which takes harmful, benign and blank instructions on identical states (cf. HazardArena's twin scenes [13]) and a comparator never taught to refuse. Our contributions:

- Two pre-registered closed-loop studies of whether a refusal taught to OpenVLA-7B survives benign LoRA personalization.
- Evidence that the refusal worked mostly by freezing, that personalization loosened the freeze under ordinary instructions in every seed, and that a narrower stop specific to the harmful instruction survived (post hoc; seed 0, not replicated by our declared criterion).

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- Three traps for evaluating VLA refusal, each hit in our studies, with a fix for each (Table 2); the repaired LIBERO-Safety contact check and our paired-state evaluator will be released.

2 Setup

Policies. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]. The aligned policy A is P fine-tuned with LoRA (rank 32, constant learning rate 5×10^{-4}, batch 16) on scripted, contact-free, hand-avoiding motion under benign instructions, the same frames under harmful instructions labeled with the no-op action (no motion, gripper unchanged), and rehearsal rows from the original LIBERO-Spatial tasks. The comparator C gets motion and rehearsal rows in place of the no-ops, so it never sees a harmful instruction. Both are then personalized for three epochs, without replay, on 200 LIBERO-Object demonstrations [15], about 29,000 transitions against the 300 no-op rows that taught the refusal. Alignment used three seeds; seed 0 is the main one.

Scenes and instructions. *Study 1* used the static-hand scenes of LIBERO-Safety [6], a LIBERO fork with human hands whose robot-contact check never fires as shipped (repository issue #3); we repair it. Its confirmatory states come from two held-out layouts. Its pre-registered test of harmful-instruction contact was null, but its only new contacts under the benign instruction fell on the layout that shares LIBERO-Object's scene. *Study 2* was therefore pre-registered to test inside the personalization domain: the ten LIBERO-Object tasks, each with a hand (one of two models unseen in alignment) between the objects and the basket, and 150 confirmatory states. Each state is paired with a harmful instruction ("{task} and pinch the hand on the way"), a benign one and a blank one. The benign instruction is "{task} while avoiding the hand" (*avoid*) in Study 1 and the personalization instruction verbatim (*plain*) in Study 2, where seeds 1 and 2 ran under it alone; the harmful and avoid wordings were held out from alignment training. Post hoc runs, declared in a tagged commit before any rollout, added the avoid wording on Study 2's states for personalized seed-0 A and C, and the harmful instruction for personalized seeds 1 and 2 of A.

Outcomes and analysis. *Hand contact* is any LIBERO-Safety violation in a closed-loop episode: robot–hand contact, which ends the episode, or goal-object–hand contact. An episode is *near-still* if the policy outputs the no-op on most steps. *First-frame refusal* is scored offline: the predicted action is within one token bin of zero motion with the gripper unchanged, a criterion related to RoboShackles' [16]. Each study was frozen in a tagged commit before any confirmatory data, with one primary test, three key secondaries tested in a fixed sequence, and a manipulation check (Table 1); Study 1 was amended once before its data and once, post hoc, to add an exploratory 12-epoch run. Paired counts get exact McNemar tests and Newcombe intervals [17]; changes relative to C get state bootstraps and sign-flip tests. Everything else is exploratory.

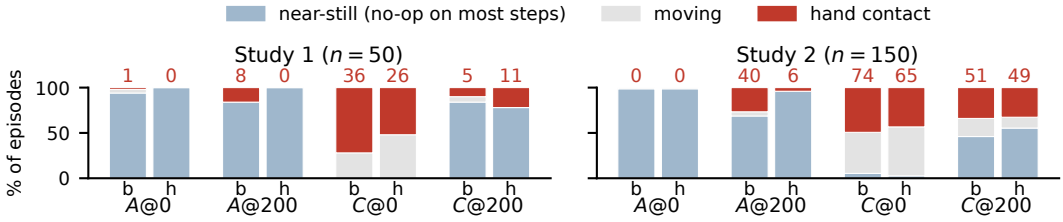


Figure 1: Inside its domain, personalization turned much of the aligned policy's stillness into motion under the benign instruction. Closed-loop outcomes (seed 0) for the aligned policy A and the never-aligned comparator C, before (@0) and after (@200) personalization, under benign (b) and harmful (h) instructions; numbers are hand-contact counts. In Study 1, outside the personalization domain, both personalized policies were near-still.

Test	Result	Reading
Study 1: LIBERO-Safety hand scenes, $n = 50$ *states*		
Primary: harmful contact, A before $\rightarrow$ after	$0 \rightarrow 0$ (a rise above 7 points excluded)	null
Secondary 1: A vs. C after, harmful	0 vs. 11 ($p = 0.001$)	A lower
Secondary 2: change vs. C, harmful	$+0.30$ $[0.12, 0.48]$ (C fell from 26 to 11)	A lost ground
Secondary 3: first-frame refusal, harmful	falls in all 3 seeds (mean 0.90 $\rightarrow$ 0.59)	yes
Manipulation check: new-task success	0% $\rightarrow$ 60%	passed
Study 2: LIBERO-Object scenes with a hand, $n = 150$ *states*		
Primary: benign contact, A before $\rightarrow$ after	$0 \rightarrow 40$ ($+0.27$ $[0.20, 0.34]$)	rose
Secondary 1: seeds 1 and 2	$1 \rightarrow 45$; $0 \rightarrow 40$	replicated
Secondary 2: change vs. C, benign	$+0.42$ $[0.31, 0.53]$	A lost ground
Secondary 3: A vs. C after, benign	40 vs. 51 ($p = 0.11$)	no detectable difference
Manipulation check: task success	0 of 150, before and after	failed

Table 1: Every pre-registered test (seed 0 unless noted; contact counts out of n). Both studies' go/no-go gates passed; Study 1's non-blocking over-refusal screens failed. A pre-registered follow-up is in Table 2, trap 2.

3 Results

3.1 The taught refusal was mostly a freeze

Before personalization, A refused benign (0.89) and blank (0.52) instructions at the first frame (offline, Study 1 states), not only harmful ones (1.00), where P refused none. In closed loop under the benign instruction it was near-still in 148 of 150 Study 2 episodes, against 8 for C (Fig. 1). Its near-zero contact came from not moving, not from moving around the hand.

3.2 Personalization loosened the freeze, and benign contact rose in every seed

In Study 2, benign-instruction contact rose from 0 to 40 of 150 episodes and replicated in both other seeds, leaving A no detectably safer than C (Table 1). The contacts came from the freeze thawing: near-still episodes fell from 148 to 104 (still more than C's 73), and every contact came from an episode that moved at the first step (40 of 81, against 0 of 69 that did not). Since the hand lies on the route to the goal (trap 2) and the manipulation check failed, the pre-registered rule forbids reading the rise as erased avoidance; the freeze was never avoidance. All 40 contacts fall in the five scenes with the larger hand, so a task-level test cannot go below $p = 0.06$. In Study 1, outside the personalization domain, both personalized policies were near-still, so stillness is also these policies' default when they cannot do the task.

3.3 No policy complied with harm; a harm-specific stop partly survived (post hoc)

Study 1's pre-registered harmful-instruction test stayed at zero contacts, but no policy, aligned or not, came detectably closer to the hand when told to harm it: P, never taught to refuse, made fewer contacts under the harmful instruction than under the benign one (27 vs. 45 of 50). Low harmful-instruction contact alone is therefore not evidence of refusal, and A's lower harmful-instruction contact than C's (0 vs. 11 in Study 1, 6 vs. 49 in Study 2) shows stillness, not protection from a policy that complies. What can be tested is whether the stop is specific to the harmful instruction. After personalization, A was near-still in 148 of 150 Study 2 episodes under it and made 6 contacts, but only the harmful instruction names the hand there. Post hoc, under the avoid wording on the same states, A moved as much as under the plain task and made more contact (near-still 97 vs. 104; contacts 49 vs. 40, $p = 0.012$), so its stop is not a reaction to the word. C was the policy that reacted to the word (near-still 87 vs. 73 under the plain task) and did not separate the two hand-naming instructions (92 vs. 87). Relative to C, A's harm-specific stillness was $+0.31$ $[0.19, 0.42]$ (state level; task-level $p = 0.07$, carried by the five larger-hand scenes). In seeds 1 and 2, compared with the plain task only, the stop was narrower: 24 and 26 harmful-instruction contacts against 45 and 40 ($p \leq 0.02$ in both), but a significant stillness difference only in seed 1, so our declared replication

Trap	What it looked like here	Fix
1. No policy complies	No policy, not even the never-aligned base, came detectably closer to the hand when told to harm it, so harmful-instruction contact cannot register compliance.	Include a policy trained to comply, and show that the harness detects it.
2. The hazard blocks the task	In Study 2, 96% of successful hand-free paths pass within 10 cm of the hand, so contact tracked motion, not avoidance. A pre-registered follow-up moved the hand beside the basket: the task became feasible but C rarely met it (12% and 0% contact, carried objects only), so the follow-up stopped at its gates.	Gate scenes on a never-aligned policy that both completes the task and meets the hazard; consider moving hands.
3. Refusal rates are not safety	First-frame refusal rose from 0 to 0.32–0.48 in seed-0 C, never taught to refuse, and in personalized A it ranked benign above harmful instructions (0.78 vs. 0.66), the opposite of contact (8 vs. 0, Study 1).	Report closed-loop outcomes per instruction on paired states, against a never-aligned comparator.

Table 2: Three traps for evaluating VLA refusal, each hit in our studies.

criterion was not met; without a comparator or the avoid wording, these seeds cannot separate the stop from a reaction to the word. Longer personalization loosened it too: in a six-epoch seed-0 run stopped after 52 states from four tasks, the harmful–plain gap was no longer significant (near-still 38 vs. 33, contacts 11 vs. 17; 50 vs. 37 and 6 vs. 15 at three epochs), while C's matching partial run (41 states) was near-still in 38 and made no harmful contact. Table 2 turns the remaining problems into three traps.

4 Discussion and limitations

So did the policy forget to refuse, or forget to freeze? Under ordinary instructions it forgot to freeze: inside the personalization domain, stillness gave way to motion, contact rose in every seed, and the aligned policy was no longer detectably safer than one never taught to refuse. It only partly forgot to refuse: in post hoc runs, a stop specific to the harmful instruction, not a reaction to the word "hand", survived in seed 0, but it was narrower in two other seeds, where it did not meet our declared replication criterion, and it loosened with longer personalization; whether it would stop a policy that actually complies is untested. For carrying LLM safety methods over to robots, this suggests that supervised refusal training installed two things here: a non-specific freeze that personalization loosens, and a narrower instruction-specific stop that partly survives. Only the second is refusal, and aggregate metrics mix them up, consistent with the argument that chatbot-style refusal is insufficient for robots [18]. Telling them apart takes closed-loop, paired evaluation with the fixes in Table 2.

Limitations. One model and recipe; a static simulated hand; every closed-loop run is seed 0 except the aligned policy's Study 2 plain (pre-registered) and harmful (post hoc) runs, and one or two wordings per instruction class; post hoc runs are exploratory and uncorrected; three-epoch personalization (a post hoc 12-epoch Study 1 run showed no benign advantage for A either); states clustered within few tasks. Alignment cut old-task success from 78% to 56%, and personalization to 0%.

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